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**SCHOOL OF ELECTRICAL ENGINEERING**

# Obstacle warning system for drivers

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**Obstacle warning system for drivers**

BY

**GROUP 3**

Under the guidance and approval of the committee, and approved by its members, this thesis has been accepted in partial fulfillment of the requirements for the degree.

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# HONESTY DECLARATION

Every member of our Group (Group 3) would like to declare that, apart from the acknowledged references, this thesis either does not use language, ideas, or other original material from anyone; or has not been previously submitted to any other educational and research programs or institutions. I fully understand that any writings in this thesis contradicted to the above statement will automatically lead to the rejection from the EE program at the International University – Vietnam National University, Ho Chi Minh City.

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# TURNITIN DECLARATION

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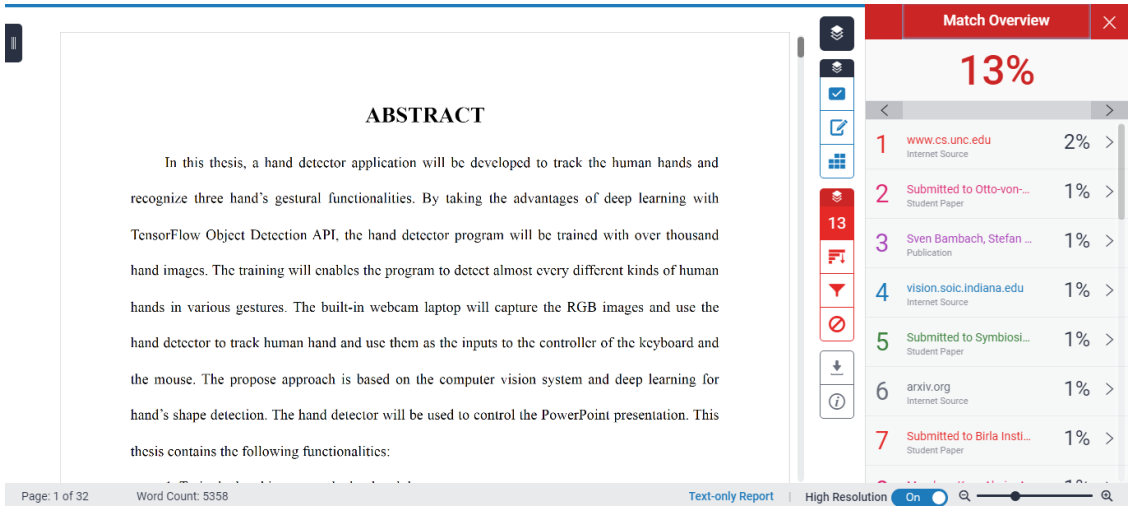

ABSTRACT

In this thesis, a hand detector application will be developed to track the human hands and recognize three hand's gestural functionalities. By taking the advantages of deep learning with TensorFlow Object Detection API, the hand detector program will be trained with over thousand hand images. The training will enables the program to detect almost every different kinds of human hands in various gestures. The built-in webcam laptop will capture the RGB images and use the hand detector to track human hand and use them as the inputs to the controller of the keyboard and the mouse. The propose approach is based on the computer vision system and deep learning for hand's shape detection. The hand detector will be used to control the PowerPoint presentation. This thesis contains the following functionalities:



Advisor Signature

Student Signature

# ACKNOWLEGMENT

We wish to convey our heartfelt appreciation to Dr. Nguyen Binh Duong, who acted as my advisor and mentor throughout the Capstone project. His meticulous guidance was essential to the project's smooth progression. Thanks to his support and motivation, we were able to successfully achieve our objectives.

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# TABLE OF CONTENTS

HONESTY DECLARATION ..... ii
TURNITIN DECLARATION ..... iii
ACKNOWLEGMENT ..... iv
TABLE OF CONTENTS ..... v
LIST OF TABLES ..... vii
LIST OF FIGURES ..... viii
ABBREVIATIONS AND NOTATIONS ..... x
ABSTRACT ..... xi
CHAPTER I INTRODUCTION ..... 1
1.1 Issue (Traffic Congestion) ..... 1
**1.1.1 The low quality of public transport services.** ..... 2
**1.1.2 Traffic laws still have many shortcomings.** ..... 3
**1.1.3 Traffic congestion significantly increases emissions** ..... 3
**1.1.4 Suffer from respiratory diseases because of polluted air** ..... 4
**1.1.5 Traffic accidents lead to long traffic jams** ..... 4
1.2. Problem statement. ..... 5
1.3 Obstacle warning system for car ..... 6
CHAPTER II LITERATURE REVIEW ..... 8
2.1 Warns the driver of signs of distraction ..... 8
2.2 Recognize the slipperiness of the road surface for vehicles ..... 9
2.3 Detects forward collision due to slippage or distraction ..... 10
2.4 System helps detect objects near the vehicle during traffic jams. ..... 11
CHAPTER III METHODOLOGY ..... 14
Goal: Develop an obstacle warning system for drivers. ..... 14
3.1 Install the ultrasonic sensor system. ..... 14
**3.1.1** Connect one ultrasonic sensor to the MCU. ..... 14
**3.1.2** Install the ultrasonic sensors on the vehicle. ..... 15
**3.1.3** Connect the laptop to the remaining sensors. ..... 15
3.2 Camera System Implementation and Image Processing ..... 16
**3.2.1** Task 1: Suitable hardware and setup. ..... 16
**3.2.2** Task 2: Image Calibration and Undistortion ..... 17
**3.2.3** Task 3: Image Stitching and Processing ..... 18

**3.2.4** Task 4: Testing and Debugging ..... 18
3.3 Sensor Integration and User Interface (UI) Development ..... 18
**3.3.1** Task 1: Ultrasonic Sensor Integration ..... 18
**3.3.2** Task 2: User Interface and apps design ..... 19
**3.3.3** Task 3: System Synchronization and Display ..... 19
3.4 Unit test and integration test ..... 19
CHAPTER IV Standard and Realistic constrain ..... 21
4.1. Standards ..... 22
4.2 Realistic Constraints ..... 23
**4.2.1** Economics ..... 23
4.2.2 Physical ..... 23
4.2.3 Environmental ..... 23
4.2.4 Ethical ..... 23
4.2.5 Reliability ..... 23
CHAPTER V PROJECT MANAGEMENT ..... 24
6.1 Cameras at each location ..... 25
6.2 The screen display all cameras ..... 27
6.3 Whole complete system ..... 28
CHAPTER VII CONCLUSION AND FUTURE WORK ..... 29
REFERENCES ..... 30
APPENDICES ..... 32

# LIST OF TABLES

Table 1. The effect of temperature on the properties of air .....12
Table 2: Unit test .....20
Table 3: Integration test .....20
Table 4 : Standard constrains and Realistic constrain of components .....22
Table 5: Device and Related Standards .....23
Table 6: PROJECT MANAGEMENT .....24

# LIST OF FIGURES

Figure 1.1 Traffic congestion in Vietnam.....2
Figure 1.1.1 The low quality of public transport services .....2
Figure 1.1.2 Traffic laws still have many shortcomings. [6] .....3
Figure 1.1.3 Traffic congestion significantly increases emissions.....4
Figure 1.1.4 Suffering from respiratory diseases because of polluted air .....4
Figure 1.1.5 Traffic accidents lead to long traffic jams [8] .....5
Figure 1.2.1 Traffic Congestion in HCMC: A Year in Review with 4,400+ Incidents at Key Locations.....5
Figure 1.2.2 Traffic congestion statics around the world .....6
Figure 2.1.1 Eyes aspect ratio .....9
Figure 2.1.2 Estimation of driver fatigue by pressure distribution on seat in long term driving.....9
Figure 2.2.1 Extended Kalman Filter for Odometry [12].....10
Figure 2.3.1 Detects forward collision due to slippage or distraction [14] .....11
Figure 2.4.1 Use bandpass filter to filter out unnecessary signal.....12

# ABBREVIATIONS AND NOTATIONS

| | |
|---|---|
| PAN: | Personal Area Network |
| MAC: | Medium Access Control |
| ADAS: | Advanced driver-assistance system |

# ABSTRACT

In this project we will develop an obstacle warning system for drivers that will help people drive their cars safer, on the other hand minimizing the amount of traffic congestion due to the bad driving behaviour of drivers when participating in traffic by detecting which roads are free to drive and strictly obey traffic signs. The car's systems will be integrated with many sensors to detect objects on the road. The driver will be easier to watch on their blind spots view to get rid of those road obstacles. It will focus most on ultrasonic sensors and camera. The car's function is easy to use, helping the driver when they sometimes can't make right decision when joining in traffic.

# CHAPTER I

# INTRODUCTION

During the process of urbanization and industrialization, traffic congestion has become increasingly serious in Vietnam, especially in major cities such as Ho Chi Minh City and Hanoi. The main causes stem from the growing number of vehicles and an underdeveloped and inefficient infrastructure system. This not only leads to prolonged delays but also causes environmental pollution and reduces work productivity.

In addition, frequent traffic congestion often leads to accidents, making the situation even more severe. According to compiled statistics, Vietnamese people spend about 2–3 hours a day commuting, resulting in billions of dollars in losses each year due to traffic jams. One of the main causes of traffic accidents and congestion is the lack of driver concentration and situational awareness, often due to fatigue or limited visibility of blind spots.

This highlights the necessity of having a system that assists drivers in observing their surroundings, especially in high-traffic environments. To support drivers in monitoring their surroundings and detecting nearby obstacles, our project aims to develop an obstacle warning system that enhances driver awareness using ultrasonic sensors installed around the vehicle and cameras. This system is capable of detecting nearby objects and accurately displaying their distance on the vehicle's screen in real time. As a result, drivers can avoid collisions and make timely decisions while participating in traffic.

## 1.1 Issue (Traffic Congestion)

More than ever before, the road network in Vietnam seems to be getting more and more crowded due to the increasing population and mobility. Road users must endure great delays due to frequent traffic congestion. This problem is shown in Figure 1.1 Traffic congestion in Vietnam cannot be ignored because it has a great impact on the development of the Vietnamese economy when workers, traders and other businesspeople have difficulty with traffic when going to work. [4]

Figure 1.1 Traffic congestion in Vietnam

### 1.1.1 The low quality of public transport services.

This section highlights the limitations of the public transport system, focusing on poor service quality, weak connectivity with supporting modes, and inadequate infrastructure—factors that hinder its expansion and efficiency.

The public passenger transport system still has shortcomings, such as the lack of connection between small-capacity transport modes such as minibuses and public bicycles to support the large-capacity public transport system. In addition, Figure 1.1.1 The low quality of public transport services, the limited infrastructure for keeping personal vehicles at stations and transfer points has directly led to difficulties in expanding the service scope of the public transport system. Subsequently, the increase in personal vehicles has also caused the public transport system to stagnate, reducing operational efficient [5]

Figure 1.1.1 The low quality of public transport services.

### 1.1.2 Traffic laws still have many shortcomings.

In this paragraph we will discuss the ongoing shortcomings in traffic laws and their implementation, emphasizing issues such as inadequate infrastructure support, confusing traffic signals, and regulations that do not align with public income levels.

After Decree 168 is shown in Figure 1.1.2 Traffic law still have many shortcomings, people's traffic awareness has improved significantly. However, the infrastructure system has not yet ensured the new regulations and there are many shortcomings. The image shows a faulty traffic light that confuses traffic participants. Following that, this new Decree is not suitable for income, leading to apathy when participating in traffic of vehicle users.

Figure 1.1.2 Traffic laws still have many shortcomings. [6]

### 1.1.3 Traffic congestion significantly increases emissions

This section addresses how traffic congestion contributes to rising emissions, highlighting issues such as prolonged engine idling, outdated infrastructure, and the growing number of fossil fuel-powered vehicles amid rapid urbanization.

Traffic congestion is shown in Figure 1.1.3 traffic congestion significantly increases emissions, slow traffic, prolonging the operating time of vehicles, engines are still running but not moving, significantly increasing emissions. Infrastructure has not been improved in parallel with the urbanization and industrialization process. Vehicle density is too high, the number of motorbikes and cars is increasing rapidly, motorbikes and other vehicles use fossil fuels instead of electric vehicles, increasing pressure on the infrastructure system and increasing emissions.

Figure 1.1.3 Traffic congestion significantly increases emissions

### 1.1.4 Suffer from respiratory diseases because of polluted air

This section explores the health impacts of air pollution caused by traffic congestion, focusing on respiratory issues, increased stress, and physical strain experienced by drivers, especially during peak hours when pollutant levels are highest.

According to many studies, frequent breathing during traffic jams and traffic congestion causes many people to suffer from chronic coughs and increased stress. In addition, Figure 1.1.4 is shown suffering from respiratory diseases because of polluted air the consequences of moving during traffic jams also seriously affect your health.Especially during rush hour, the concentration of pollutants in the air increases, breathing during this time causes drivers to suffer from respiratory diseases.Traffic jams cause drivers to sit still in one position for a long time, leading to damage to the neck and spine. [7]

Figure 1.1.4 Suffering from respiratory diseases because of polluted air

### 1.1.5 Traffic accidents lead to long traffic jams

This section examines how traffic accidents, often resulting from violations like speeding or unsafe driving, are a major cause of prolonged traffic jams, as illustrated by a severe incident on Phu My Bridge that led to a 10km backlog.

One of the reasons for long traffic jams is traffic accidents caused by traffic violations is shown in Figure 1.1.5 traffic accidents lead to long traffic jams. Violations such as speeding, going the wrong way, and not keeping a safe distance from the vehicle in front can all lead to traffic accidents. This is an image of an accident on Phu My Bridge that caused a traffic jam 10km behind it.

Figure 1.1.5 Traffic accidents lead to long traffic jams [8]

**1.2. Problem statement.**

According to a report from the Institute of Transport Strategy and Development is shown in Figure 1.2.1 Traffic Congestion in HCMC: A Year in Review with 4,400+ Incidents at Key Locations. Because of traffic congestion, Vietnamese people spend about 2 to 3 hours a day traveling. In 2022, the Department of Transport said that Ho Chi Minh City lost about 6 billion USD/year and Hanoi 3 billion USD/year because of traffic congestion. Of which, wasted fuel costs up to 30%. [1]

Figure 1.2.1 Traffic Congestion in HCMC: A Year in Review with 4,400+ Incidents at Key Locations

Traffic congestion also causes significant damage globally. According to INRIX is shown in Figure 1.2.2, in the US, the average driver will spend 43 hours commuting in 2024 and will cost the country $74 billion. In the UK, it will be £7.7 billion. [2]

In addition to economic and time losses, traffic congestion also causes other negative impacts such as slowing the movement of goods, increasing inflationary pressures, and negatively affecting the environment. All of this reduces the quality of life

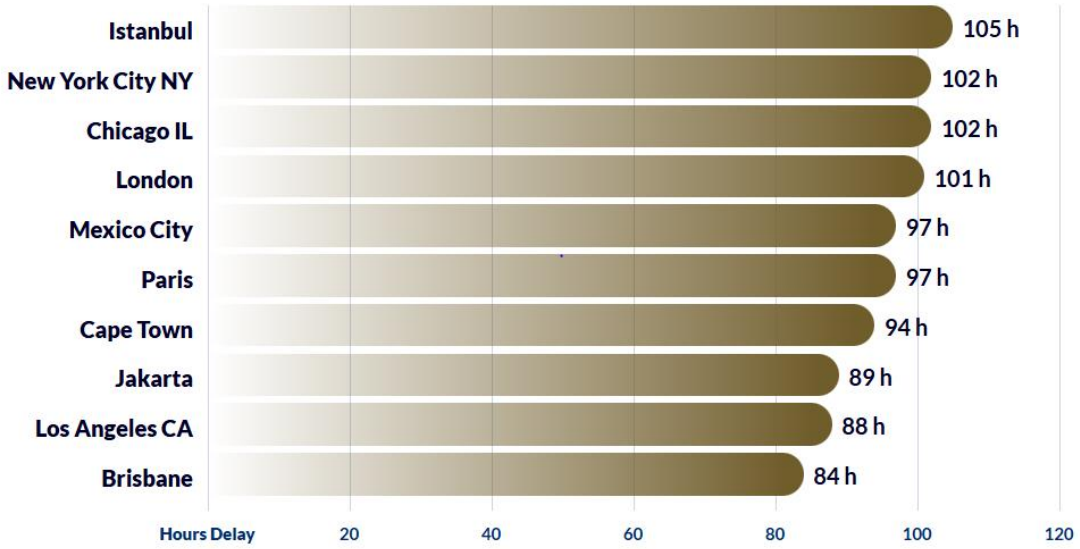



Figure 1.2.2 Traffic congestion statics around the world

1.3 Obstacle warning system for car

The objective that our group is towards is to help drivers move easily on the road and avoid having collision while driving. When you participate in traffic, there are always cars around you, leading to difficulty in recognizing the traffic situation around your car.

Objective 1: Install the ultrasonic sensor system.
Task 1: Connect one ultrasonic sensor to the MCU.
(Test 1: Check the distance and field of view (FOV) assigned to the sensor.)

Task 2: Install the ultrasonic sensors on the vehicle.
(Test 2: Check whether the distance and FOV cover all areas around the vehicle.)

Task 3: Connect the MCU to the remaining sensors.
(Test 3: Check whether the sensors send alerts to the MCU when an object is detected within the observation range (30–50 positions around the vehicle).)

Objective 2: Install the camera system.
Task 1: Connect one camera to the MCU along with the ultrasonic sensor.
(Test 4: Check whether the MCU can display images from the camera while simultaneously receiving signals from the ultrasonic sensor.)

Task 2: Install the remaining cameras at blind spots around the vehicle.
(Test 5: Check whether the entire blind spot area around the vehicle is fully covered by the cameras.)

Task 3: Synchronize the camera system with the sensor system to issue alerts when an object is detected.
(Test 6: Check whether the system displays images along with appropriate warnings when an obstacle appears in the detection area.)

Objective 3: Install the display system.
Task 1: Connect the MCU to a buzzer.
(Test 7: Check whether the buzzer emits sound when sensors detect an obstacle around the vehicle.)

Task 2: Connect a display screen to the MCU.
(Test 8: Check whether the screen shows the distance and FOV covering all positions around the vehicle, and whether it displays images transmitted from the camera.)

Task 3: Connect the entire system.
(Test 9: Check whether the buzzer emits sound and the camera image appears on the screen when an obstacle is placed around the vehicle and detected by the sensors.)

Objective 4: Unit test and integration test

# CHAPTER II

# LITERATURE REVIEW

## 2.1 Warns the driver of signs of distraction

This section addresses advanced driver monitoring systems designed to detect signs of distraction or fatigue. It introduces technologies like infrared cameras for tracking facial and eye movements using the eyes aspect ratio, and pressure sensors embedded in the seat to monitor body posture. These systems use computer vision and deep learning to analyze driver behavior and issue alerts when signs of drowsiness or inattention are detected, enhancing safety during long-term driving.

People install infrared cameras mounted on the steering wheel or near the dashboard to track the driver's face and eye movements even in a low light or at night. The blink rate is calculated by tracking the opening/closing of the eyelids through each frame of the video. [9]A common index is *eyes aspect ratio* is shown in figure 2.1.1 Estimation of driver fatigue by pressure distribution on seat in long term driving, which is used to calculate the distance between the characteristic points around the eyes. If this index decreases over a certain period, the system will count it as a blink. When measuring the blink rate over a period, if this index decreases or increases rapidly, the system will warn the driver. When capturing the driver's face, people use computer vision to determine the location of their face using the face detection algorithm to know the location of the face through each frame. Then extract important data such as eyes, nose, and mouth. Later, when I went deeper into the study, I saw that people used deep learning to determine facial movements. Specifically, a convolutional neural network is trained with image data of drivers in different states such as drowsiness, loss of concentration to assess the driver's concentration level. This is only knowing the driver's condition but not knowing the level of fatigue or sleepiness. [10] Figure 2.1.2 Estimation of driver fatigue by pressure distribution on seat in long term driving. I found that people attach pressure sensors under the car seat, people who are tired or lose concentration often tend to lean or throw their shoulders, accidentally creating a force on the seat, the sensor will measure and consider the level of fatigue. If this level exceeds the allowable level leading to risks when participating in traffic, the system will warn through the display screen, vibrate the steering wheel or warn through sound so that the driver knows to adjust his behavior and driving speed

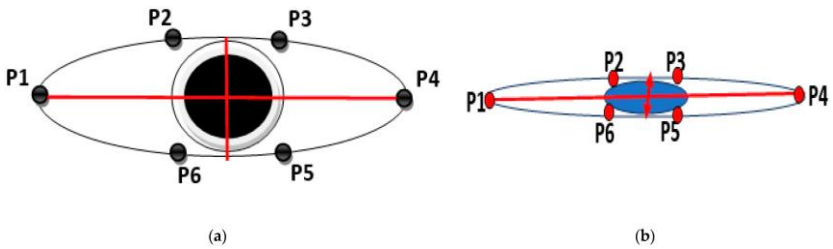


Figure 2.1.1 Eyes aspect ratio

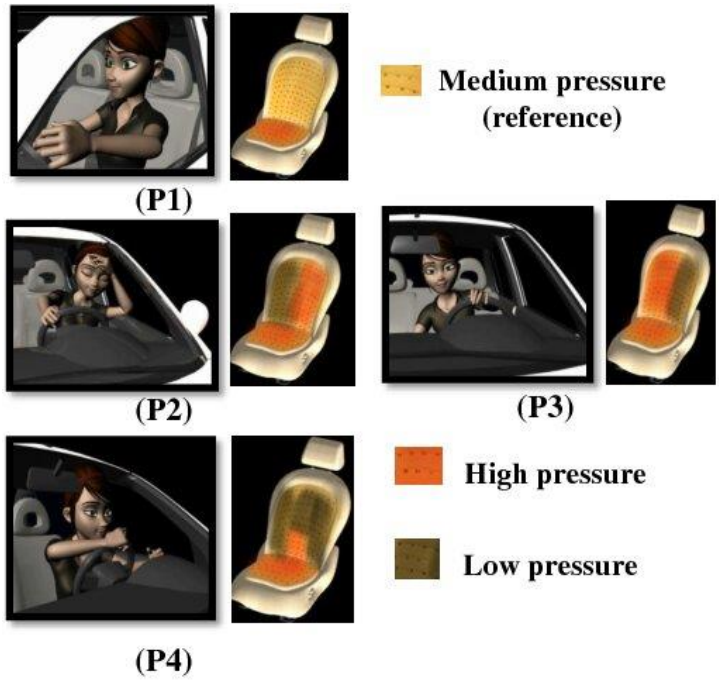


Figure 2.1.2 Estimation of driver fatigue by pressure distribution on seat in long term driving

Several technologies can be applied to enhance a 360-degree camera system with object detection around the vehicle. Specifically, an infrared camera can be integrated to monitor the driver's eyes and face using the Eyes Aspect Ratio (EAR) to detect signs of drowsiness. Additionally, pressure sensors embedded in the seat can detect abnormal posture changes caused by fatigue. Technologies such as computer vision and deep learning (e.g., CNNs) can analyze driver behavior and issue warnings. These alerts can be displayed through the 360-degree camera interface, improving safety by monitoring both surroundings and driver attention.

## 2.2 Recognize the slipperiness of the road surface for vehicles

This section introduces the use of an IMU sensor attached to the wheel axle, which measures linear, centrifugal, and original acceleration during vehicle movement. Using this data, the Extended Kalman Filter algorithm estimates the vehicle's position and speed while also detecting road conditions like wet or slippery surfaces. When slippage is detected, the system alerts the driver to adjust behavior for safer driving.

This IMU sensor is attached to the axle of the wheel. When the vehicle moves, this sensor measures three main values is shown in figure 2.2.1 Extended Kalman Filter for Odometry The first value is linear acceleration, which is the change in speed over time on a straight line, the second value

is centrifugal acceleration, which is the acceleration that only appears when the vehicle moves along a curve. The third value is the original acceleration, which is the change in the original velocity over time, if your vehicle moves faster or slower, the original acceleration will appear
When these three-input data are obtained, the Kalman filter algorithm is used, a measurement method used for robotics and vehicle navigation to help determine the position and velocity of the system through sensor data
When information about the road surface is obtained, the Kalman filter will help find different optimal boundaries between road surfaces, such as wet roads, dry roads, and road surface slipperiness. When the vehicle is moving on a wet road surface, the wheels tend to rotate faster than the actual speed of the vehicle, leading to wheel slippage. If this index exceeds the allowable threshold, the system will notify the driver to adjust behavior such as not swerving or reducing speed. [11]

$$\begin{bmatrix} \ddot{p} + \ddot{\Theta} \cdot R_s \cdot \cos\Theta + \dot{\Theta}^2 \cdot R_s \cdot \sin\Theta \\ -\ddot{\Theta} \cdot R_s \sin\Theta + \dot{\Theta}^2 \cdot R_s \cdot \cos\Theta \end{bmatrix} = \begin{bmatrix} a_{x,senzor} \\ a_{y,senzor} \end{bmatrix}, \quad (1)$$

Figure 2.2.1 Extended Kalman Filter for Odometry [12]

When wheel slippage is detected—such as when the wheels rotate faster than the actual speed of the vehicle on wet surfaces—the system will alert the driver through the camera interface, helping them adjust their behavior by reducing speed or avoiding sharp turns. Integrating this data enhances situational awareness and driving safety under complex road conditions.

**2.3 Detects forward collision due to slippage or distraction**

This section explains how a 360-degree camera mounted on the car roof captures surrounding information, such as pedestrians and vehicles, to prevent collisions. It uses the You Only Look Once (YOLO) algorithm, as shown in Figure 2.3.1, which divides the image into cells and detects objects in real-time. Each cell predicts object presence, location, and type with a confidence score, allowing the system to identify and filter potential collision risks effectively.
People mount a 360-degree camera on the roof of the car to record information such as passersby, obstacles or cars in front. Once this input information is available, people use the You Only Look Once algorithm is shown in Figure 2.3.1 Detects forward collision due to slippage or distraction, which is an algorithm that helps identify objects in the frame only once.
After having the frame image from the camera, this algorithm will divide the image into equal cells, these cells are responsible for identifying objects with a center in that cell. Each cell will make a prediction including the coordinates of that cell, the level of confidence in identifying the object in that cell and the type of object.
This confidence index indicates the possibility of an object in that cell and the accuracy of the prediction. If there is no object in the cell, this value will be zero, if there is, it will be another high value. Then filter duplicate cells to retain the best results [13]

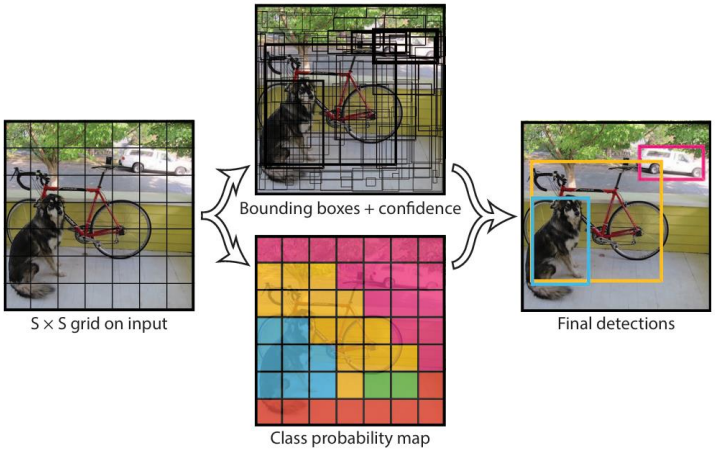



Figure 2.3.1 Detects forward collision due to slippage or distraction [14]

Integrating the YOLO (You Only Look Once) algorithm into a 360-degree camera system significantly enhances object detection around the vehicle. YOLO enables real-time identification of pedestrians, vehicles, and obstacles by dividing each frame into grids and predicting object presence, location, and type with confidence scores. This allows the system to highlight high-risk objects and filter out duplicates, ensuring accurate results. With YOLO, the 360-degree camera can not only display surrounding views but also understand what is present in each area, helping detect collision risks early and improving driver awareness and safety in dynamic environments such as traffic or parking.

**2.4 System helps detect objects near the vehicle during traffic jams.**

This section discusses the use of ultrasonic sensors to assist drivers in navigating traffic jams by detecting nearby obstacles. Typically, 8–16 sensors are installed depending on the vehicle type, providing accurate distance measurements within 5 meters. However, since ultrasonic performance is affected by temperature and certain materials, a temperature sensor is added to correct measurement errors, and a bandpass filter is used to eliminate irrelevant signals for more precise detection, as shown in Figure 2.4.1.

Traffic jams make it difficult for drivers to observe their surroundings. Our group found that people install ultrasounds around strategic locations of the vehicle, depending on the type of vehicle, tall or short, long or short, 8-16 ultrasounds will be installed. These ultrasounds work best within 5m, when measuring the distance from the vehicle to the nearest object, it emits ultrasonic waves to measure and when there are results, the camera will display the location at that place to help the driver easily observe. But when going deeper into the research, Table 1. The effect of temperature on the properties of air, it was discovered that this ultrasound is easily interfered by temperature, so people install a temperature sensor next to the sensor and use the formula between velocity and temperature to calculate the error. This ultrasound is shown in Figure 2.4.1 use bandpass filter to filter out unnecessary signal also erroneous by easily absorbing objects such as human skin or clothing, then people will use a bandpass filter to remove unnecessary small signals or large signals that do not match the parameters of the ultrasound to help determine the most accurate distance. [12]

Tác động của nhiệt độ lên các tính chất của không khí

| Nhiệt độ $T$ (Celsius\|°C) | Tốc độ âm thanh $c$ (m/s) | Khối lượng riêng của không khí $\rho$ (kg/m$^3$) | Trở kháng âm thanh đặc trưng riêng $z_0$ (Pa·s/m) |
|---|---|---|---|
| 35 | 351,88 | 1,1455 | 403,2 |
| 30 | 349,02 | 1,1644 | 406,5 |
| 25 | 346,13 | 1,1839 | 409,4 |
| 20 | 343,21 | 1,2041 | 413,3 |
| 15 | 340,27 | 1,2250 | 416,9 |
| 10 | 337,31 | 1,2466 | 420,5 |
| 5 | 334,32 | 1,2690 | 424,3 |
| 0 | 331,30 | 1,2922 | 428,0 |
| −5 | 328,25 | 1,3163 | 432,1 |
| −10 | 325,18 | 1,3413 | 436,1 |
| −15 | 322,07 | 1,3673 | 440,3 |
| −20 | 318,94 | 1,3943 | 444,6 |
| −25 | 315,77 | 1,4224 | 449,1 |

Table 1. The effect of temperature on the properties of air

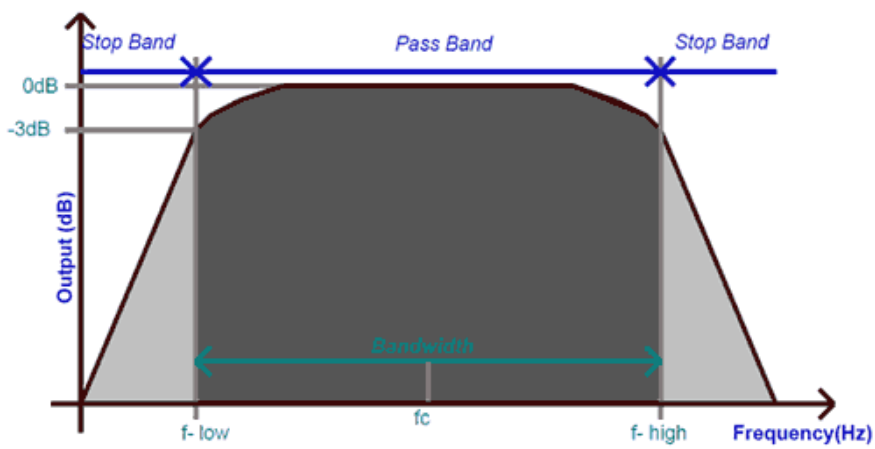



Figure 2.4.1 Use bandpass filter to filter out unnecessary signal

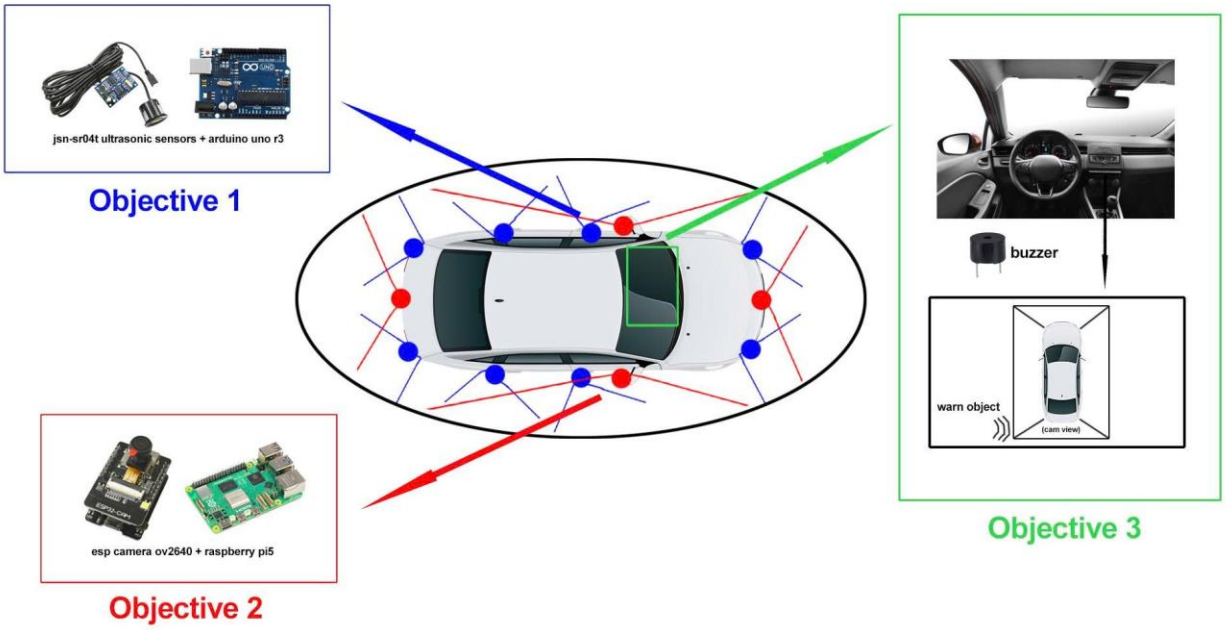
I apply the use of **ultrasonic sensors** from the section to improve my 360-degree camera and object detection system. By installing 8–16 ultrasonic sensors around the vehicle, I can measure the distance to nearby objects within 5 meters, helping drivers detect obstacles in traffic or tight spaces

**Technologies we can use for a 360-Degree Driver Assistance and Obstacle Detection System**
We use 360-degree cameras placed on sides of the car. These cameras can record everything around the vehicle.
The video is sent to a computer that uses an object detection algorithm called YOLO (You Only Look Once). This algorithm is fast and helps detect people, cars, and other objects in real time.
We also use ultrasonic sensors placed around the car. These sensors send out sound waves and measure how far objects are. They work well within 5 meters. However, ultrasonic signals can be affected by temperature. So, we add temperature sensors and use a formula to fix the errors. We also use a bandpass filter to remove bad signals and improve accuracy. In traffic jams or small

streets, these sensors are very helpful. The driver may not see people or objects close to the car, but our system shows the exact location on a screen and gives warnings by sound or vibrations.
In the future, we can add more features like monitoring the driver's face and eyes using a camera to check if they are tired or not paying attention. We can also use pressure sensors under the seat to know if the driver is sleepy based on how they move.
In conclusion, our system combines cameras, sensors, and smart software to make driving safer. It gives drivers better awareness of their surroundings and helps them avoid accidents.

# CHAPTER III

# METHODOLOGY

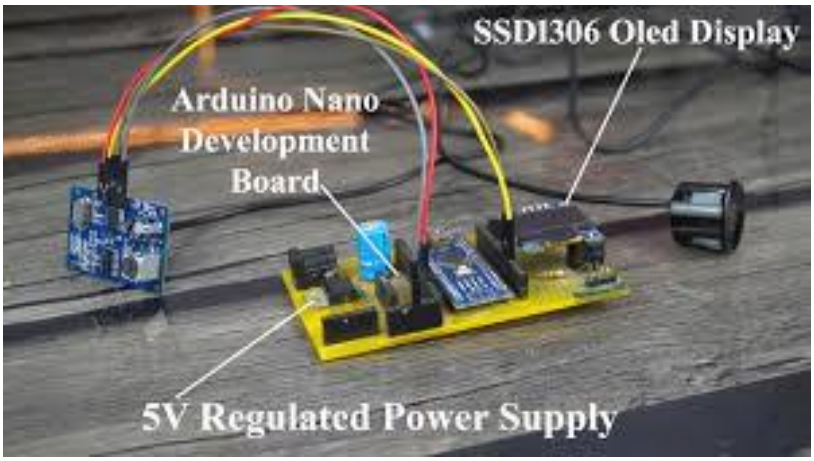



**Goal:** Develop an obstacle warning system for drivers.

**3.1** Install the ultrasonic sensor system.

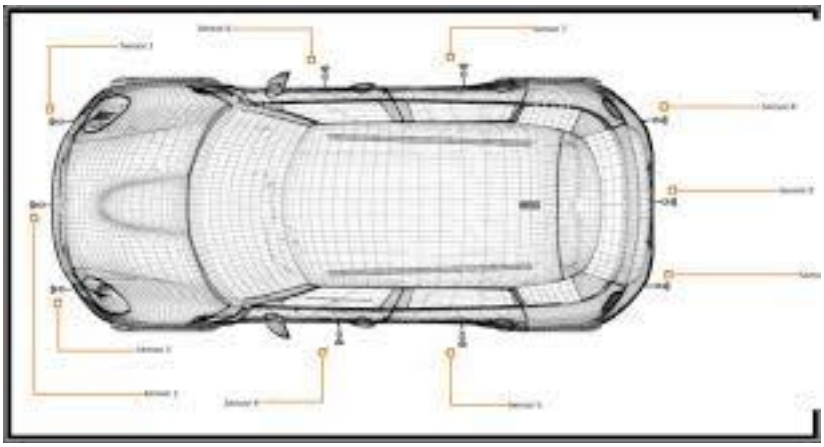
**3.1.1** Connect one ultrasonic sensor to the MCU.

The JSN-SR04T is a waterproof ultrasonic sensor that works with an Arduino Uno by sending and receiving ultrasonic pulses to measure distance. The Arduino sends a trigger pulse to the sensor, which then emits ultrasonic waves. If an object is within range, the waves reflect back, and the sensor measures the time it takes for the echo to return.




(**Test 1:** Check the distance and field of view (FOV) assigned to the sensor.)
Measure distances to objects at various angles and ranges. Verify the sensor's accuracy and check its effective field of view (FOV). Ensure consistent readings within expected angles (typically 15–30°).

**3.1.2** Install the ultrasonic sensors on the vehicle.
Mount multiple ultrasonic sensors around the vehicle: front, rear, sides, and corners. Placement should avoid blind spots and interference from the vehicle's body. Consider sensor height and tilt angle for optimal coverage.



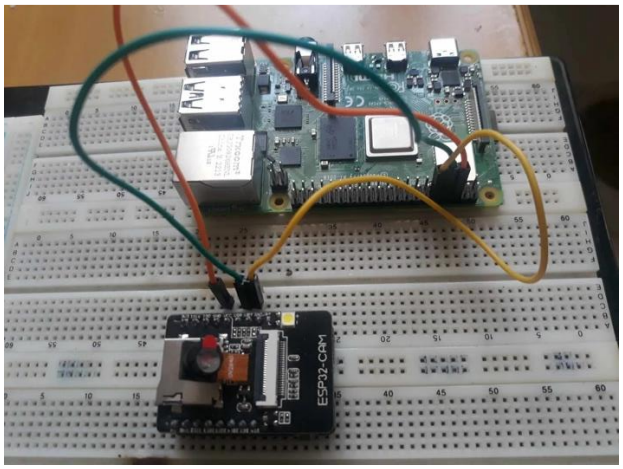
(**Test 2:** Check whether the distance and FOV cover all areas around the vehicle.)
Place objects around the vehicle and verify that all zones are detected. Ensure there are no blind spots or redundant overlaps. Test in different lighting and environmental conditions to ensure reliability.

**3.1.3** Connect the laptop to the remaining sensors.
Wire all remaining sensors to the MCU using available I/O or expanders (e.g., I2C). Update the code to handle multiple sensors, gather real-time distance data, and process alerts when objects are nearby.

(**Test 3:** Check whether the sensors send alerts to the MCU when an object is detected within the observation range (30–50 positions around the vehicle).)
Simulate objects at 30–50 positions around the vehicle. Check if each sensor correctly detects nearby objects and sends alerts to the MCU. Verify alert timing and accuracy for real-time driver feedback.

**3.2** Camera System Implementation and Image Processing

**3.2.1** Task 1: Suitable hardware and setup.




- **Lens Camera:** Choose a camera with a fisheye lens to be able to monitor narrow spaces and wide angles. Required specifications:A resolution of 1280x1280 or 720p with a frame rate of 30fps ensures clear images and real-time playback.

  This is the picture and information of the cameras that we will use in our system.

- **Camera hardwares:** It doesn't need to be the best, but it must at least be good enough to process data quickly, minimize latency, and the camera should use a USB 2.0 or 3.0 connector.

  Since we don’t have any good MCUs as they are too expensive, we will connect the camera to our personal computer to keep its stability throughout time.

**3.2.2** Task 2: Image Calibration and Undistortion

- **Parameter Extraction:** Perform the calibration process to extract the parameters. From the original image, calculate the K coefficient (Camera Matrix) and the D coefficient (Distortion Coefficient).

  The figure below illustrate how we extract the K-D parameters for further task.

- **Undistortion Processing:** Using the K and D parameters found, program an algorithm to reduce image distortion, refining the curved fisheye image into a flat image or an image with a more realistic aspect ratio for easier observation and processing.

  The picture we show illustrate that the K-D parameters act a very important role in the process.

**3.2.3** Task 3: Image Stitching and Processing

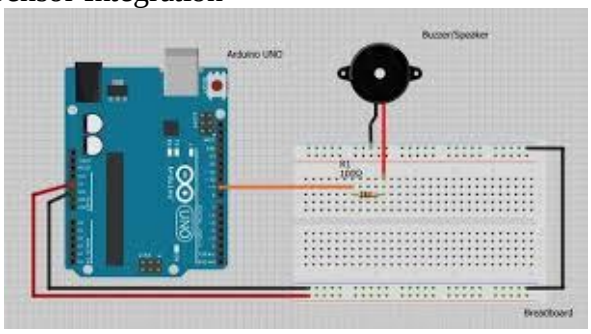
- **View Merging:**The process involves stitching or arranging processed camera footage to create a continuous view or frames that match the driver's perspective.
- **Optimization for Raspberry Pi 5:** Optimize the code to match the processing speed of the Raspberry Pi 5 CPU. Ensure the image processing algorithm runs stably at 30fps without overloading the system.

**3.2.4** Task 4: Testing and Debugging

- Conduct testing to identify errors in image processing and data transmission.
- It is expected that approximately 20 errors related to lag or frame drops will be identified and fixed to further optimize system performance.

**3.3** Sensor Integration and User Interface (UI) Development

**3.3.1** Task 1: Ultrasonic Sensor Integration



- **Hardware Connection:** Connecting ultrasonic sensors directly to the GPIO pins of the Raspberry Pi 5 reduces hardware complexity and increases signal processing speed.

- **Distance Measurement Logic:** Develop an algorithm to read signals from GPIO to calculate the distance to an obstacle. Perform several real-world measurement tests to adjust for errors.

**3.3.2** Task 2: User Interface and apps design

- **Interface Coding:**The user interface (GUI) should be programmed to display on the screen. The interface should be easy for the driver to see so they can focus on driving.

- **Apps design:** Draw recognizable icons to represent the vehicles and warnings for use in the application.

**3.3.3** Task 3: System Synchronization and Display

- **Data Visualization:** Integrate images from the camera and measured distances into the interface.

- **Display:** The screen displays the distance measured by the ultrasonic sensor and the image from the camera directly.

**3.4** Unit test and integration test

| Unit Test | Description | Condition | Check |
|---|---|---|---|
| 1 | Check the distance and field of view (FOV) assigned to the sensor | The distance and field of view (FOV) assigned to the sensor | Successfull |
| 2 | Check whether the distance and FOV covered the back of the vehicle | The distance and FOV covered the back of the vehicle | Successfull |
| 3 | Check whether the code can export to display images from the camera while simultaneously receiving signals from the ultrasonic sensor | The code can export to display images | Successfull |

| | | | |
|---|---|---|---|
| 4 | Check whether the entire blind spot area around the vehicle is almost completely covered by the cameras | Blind spot area around the vehicle is almost completely covered by the cameras | Successfull |
| 5 | Check whether the alert on screen when sensors detect an obstacle around the vehicle | The alert on screen | Successfull |

Table 2: Unit test

| **Integration Test** | **Description** | **Result** | **Date** |
|---|---|---|---|
| 1 | Check whether the sensors send alerts to the laptop when an object is detected within the observation range (30–50 positions around the vehicle) | The sensors send alerts to the laptop | Successfull |
| 2 | Check whether the system displays images along with appropriate warnings when an obstacle appears in the detection area | The system displays images | Successfull |
| 3 | Check whether the screen shows the distance and FOV covering all positions around the vehicle, and whether it displays images transmitted from the camera | The screen shows the distance and FOV covering all positions | Successfull |
| 4 | Check whether the alert on screen and the camera image appears on the screen when an obstacle is placed around the vehicle and detected by the sensors | The alert on screen and the camera image appears on the screen | Successfull |

Table 3: Integration test

# CHAPTER IV Standard and Realistic constrain

In this chapter, we will provide the standards and realistic constraints that were employed in this Thesis project, as well as the hardware and software that was used.

• ASME Standards
ASME Y14.5 – Geometric Dimensioning and Tolerancing

• ISO Standards
ISO 9001 – Quality Management Systems

• IEEE Standards
IEEE 802.11 a/b/g/n/ac/ax – WiFi Standards
IEEE 802.15.1 – Bluetooth

| Number | Component | Standard constrains | Realistic constrain |
|---|---|---|---|
| 1 | Raspberry pi 5 | Operating Voltage: 5V DC<br>Current Consumption: ~3A (under high load)<br>Operating System: Raspberry Pi OS (Linux)Supports GPIO, USB, WiFi, Bluetooth | Operating temperature may rise significantly, requiring heat dissipation<br>Depends on a stable power supply<br>SD card may get damaged if power is suddenly cut<br>High cost and limited availability on the market |
| 2 | Arduino uno r3 | Operating Voltage: 5V<br>Input Voltage via USB or Jack: 7–12V<br>14 Digital I/O Pins (6 PWM)<br>6 Analog Input Pins | Limited memory, making it difficult to run complex algorithms<br>No built-in WiFi/Bluetooth<br>Does not support multitasking<br>Easily damaged if powered incorrectly |
| 3 | JSN-SR04T V3.0 Waterproof Probe Ultrasonic Sensor | Operating Voltage: 5V<br>Measuring Range: 20cm – 600cm<br>Frequency: 40kHz | Accuracy decreases in noisy or windy environments, or with sound-absorbing materials |

| | | | |
|---|---|---|---|
| | | Communication Interface: TTL | Minimum distance of 20cm makes close-range measurement difficult<br>Long cables can cause signal attenuation |
| 4 | ESP32-CAM with OV2640 | Operating Voltage: 3.3V – 5V<br>Camera with up to 2MP resolution<br>Built-in WiFi and Bluetooth<br>No built-in USB port for uploading code | Heats up during prolonged operation<br>Requires a USB-TTL converter module to upload code<br>WiFi may experience interference if poorly designed<br>Limited RAM for image processing |
| 5 | Buzzle | Operating Voltage: 3V – 5V<br>Current Consumption: a few mA<br>Available in active and passive types | Sound may be low depending on the type<br>Can be annoying if used continuously<br>Easily damaged if supplied with incorrect voltage |
| 6 | LM2596 | Input Voltage: 6V – 24V (depending on the type)<br>Output Voltage: stable 5V<br>Maximum Current: 2A–3A | May overheat under heavy load<br>Poor-quality units may cause electrical noise<br>Lower efficiency with small loads or low input voltage |

Table 4 : Standard constrains and Realistic constrain of components

4.1. Standards

ASME Standards

- ASME Y14.5 – Geometric Dimensioning and Tolerancing

ISO Standards

- ISO 9001 – Quality Management Systems
- ISO 14001 – Environmental Management

- ISO/IEC 27001 – Information Security Management

IEEE Standards

- IEEE 802.11 a/b/g/n/ac/ax – WiFi Standards
- IEEE 802.15.1 – Bluetooth

| Device | Standards |
|---|---|
| Wifi module | IEEE 802.11 a/b/g |
| Bluetooth module | IEEE 802.15.1 or 802.15.4 |
| Zigbee module | IEEE 802.15.4 |
| Drone Frame F450 | ASME B89 (Dimensional Accuracy), ISO 2768 |
| Camera Sensor | IEEE 2030 (Smart Systems) |
| Power module | IEEE 1547 (Power Interfacing) |

Table 5: Device and Related Standards

## 4.2 Realistic Constraints

**4.2.1** Economics

Cost: around 150 USD to 300 USD.

4.2.2 Physical

Size: Approx. 100 mm × 60 mm × 40 mm per module (e.g., for one sensor node or camera).
Weight: <3kg.

4.2.3 Environmental

Noise pollution: <70dB
Soil pollution: None during usage; recyclable electronics if properly disposed.

4.2.4 Ethical

The 360° Driver Assistance System strictly avoids unauthorized surveillance and respects user privacy.

4.2.5 Reliability

General performance: The system provides real-time 360° situational awareness and object detection within a 0.2 m to 6 m range.

# CHAPTER V

# PROJECT MANAGEMENT

| | | | |
|---|---|---|---|
| Objective 1 | | | |
| Task 1 | 2-Jun-25 | 4-Jun-25 | Nghi + Quan |
| Task 2 | 4-Jun-25 | 6-Jun-25 | Nghi + Quan |
| Task 3 | 6-Jun-25 | 8-Jun-25 | Nghi + Quan |
| | | | |
| | | | |
| Objective 2 | | | |
| Task 1 | 11-Jun-25 | 12-Jun-25 | Dai + Qui |
| Task 2 | 12-Jun-25 | 14-Jun-25 | Dai + Qui |
| Task 3 | 14-Jun-25 | 15-Jun-25 | Dai + Qui |
| | | | |
| | | | |
| Objective 3 | | | |
| Task 1 | 17-Jun-25 | 19-Jun-25 | Nghi + Khoa |
| Task 2 | 19-Jun-25 | 20-Jun-25 | Nghi + Khoa |
| Task 3 | 20-Jun-25 | 21-Jun-25 | Nghi + Khoa |
| | | | |

Table 6: PROJECT MANAGEMENT

# CHAPTER VI
# EXPECTED RESULTS

In this part, the result of our project - Obstacle warning system for drivers.

## 6.1 Cameras at each location

These photos below show the front, behind and both left/right camera of the car displaying individually.

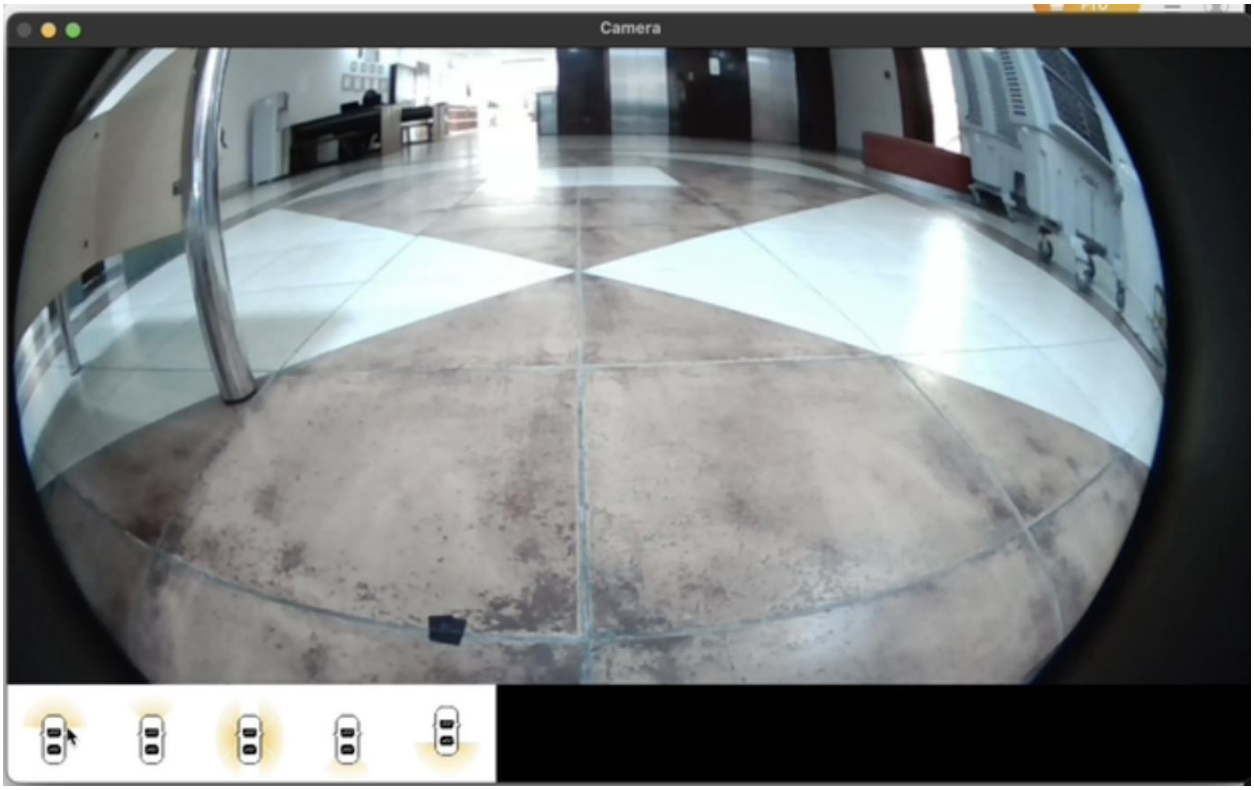



Figure 6.1.1 The front camera

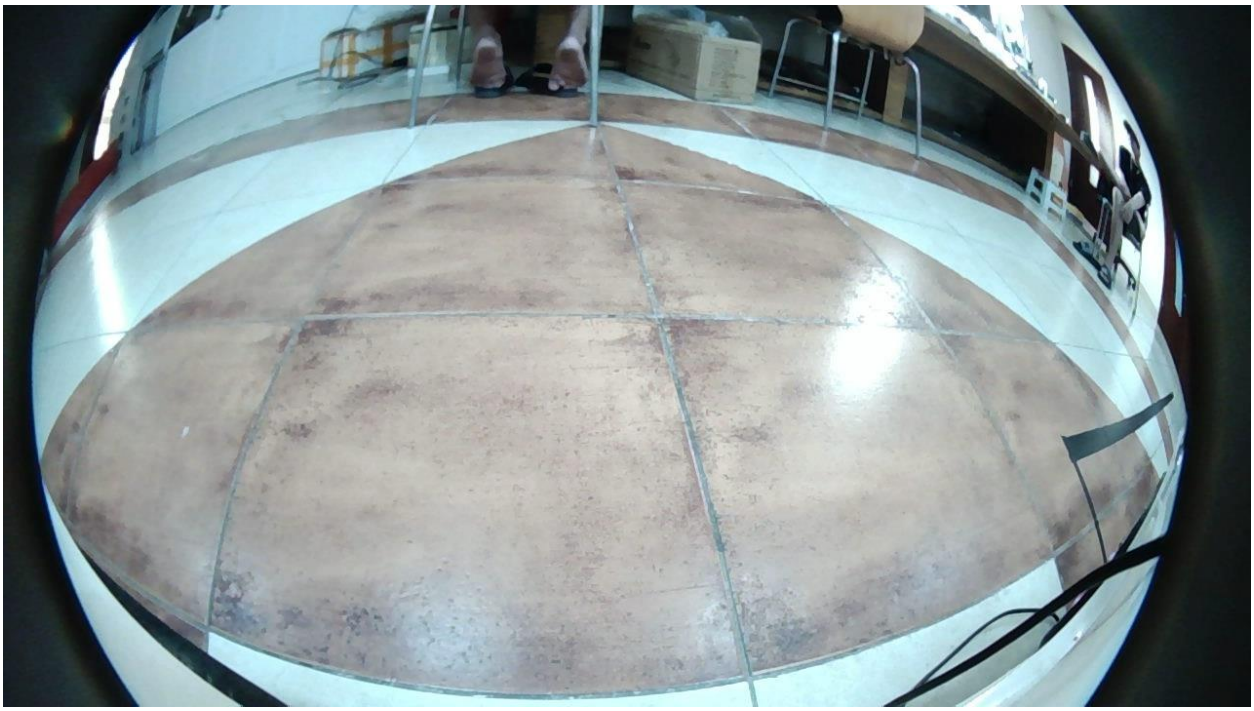


Figure 6.1.2 The right camera

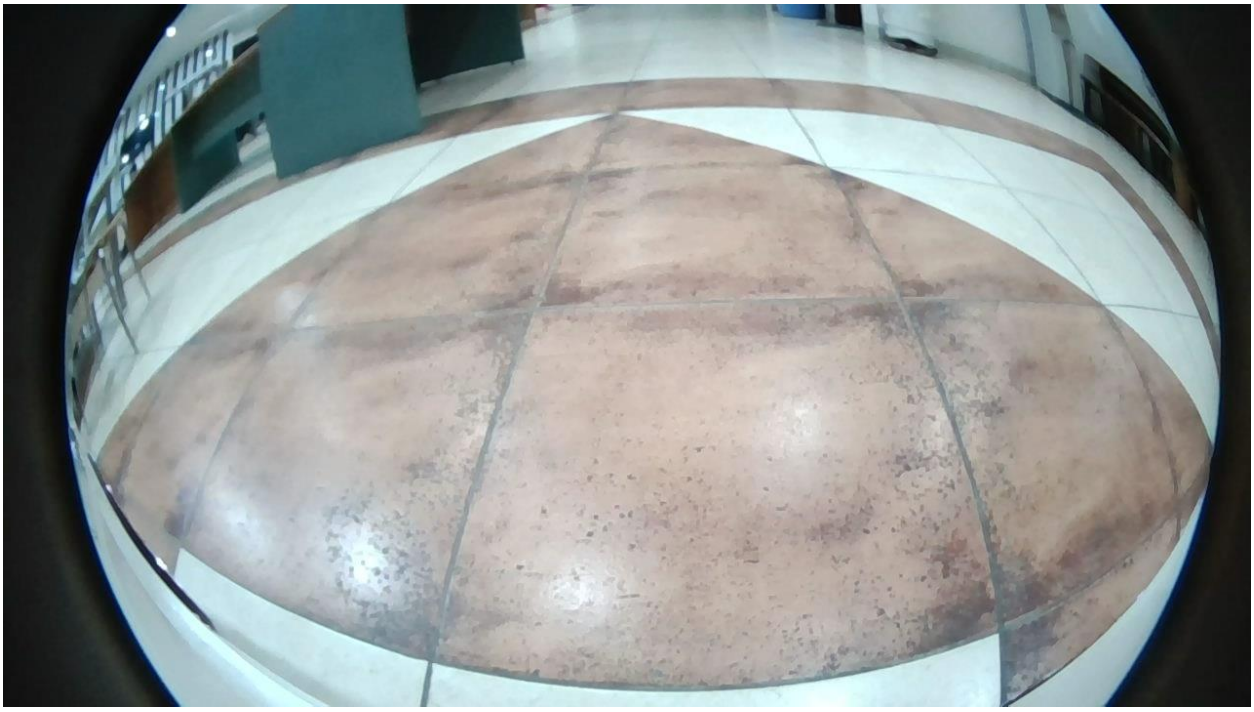


Figure 6.1.3 The left camera

Figure 6.1.4 The behind camera

## 6.2 The screen display all cameras

The picture below show us the camera system after calibrating and stitching.

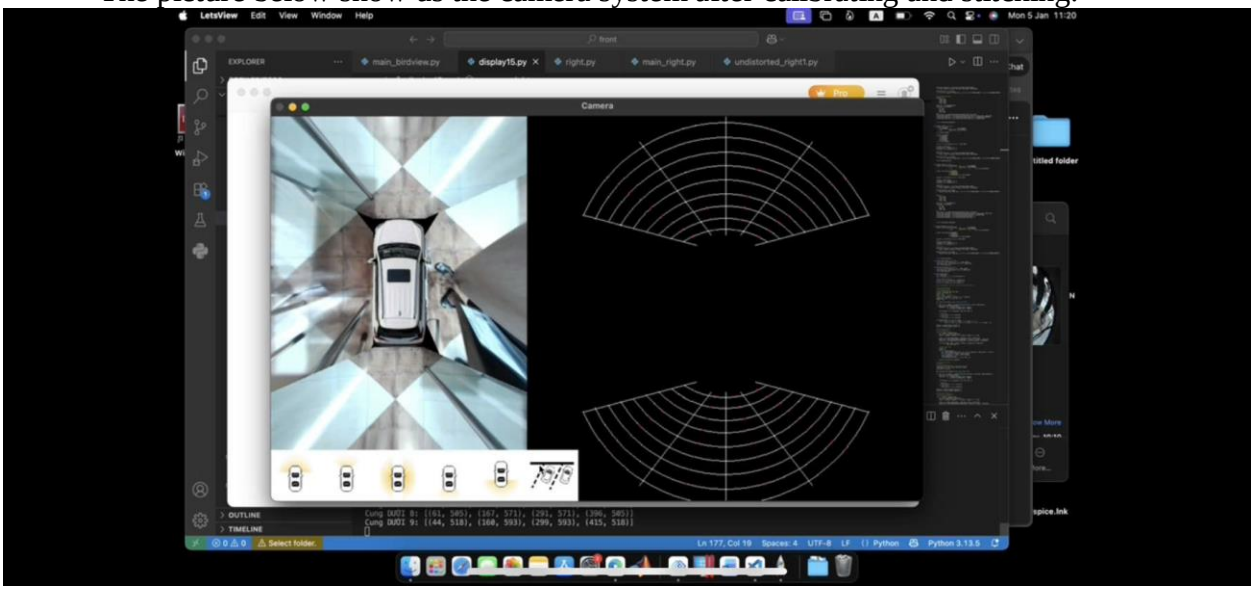



Figure 6.2.1 Whole camera showing

### 6.3 Whole complete system

Figure 6.3.1 The implemented system

# CHAPTER VII

# CONCLUSION AND FUTURE WORK

## 7.1 Conclusion

This project aims to design an obstacle warning system for drivers by cameras to detect potential hazards around the vehicle. The system provides timely visual to assist drivers in recognizing obstacles early, thereby reducing the risk of collisions and improving overall driving safety.

## 7.2 Future work

For more safety , this project should include the car warning sound ( horn ) and warning lights to warn the driver enhance safety and minimize unecessary risks .

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# APPENDICES